Mid term review

Exercice 1 : TCP/ UDP

Suppose you are designing the communication part of a multi-functional teleconferencing app. Your teleconferencing app allows voice talk and text-based messaging among the users. The table below shows the QoS (quality of service) the general users demand for the two functions.

	Loss	Bandwidth	Time Sensitive
Messaging	no loss	elastic	no
Talk	loss tolerant	audio: 5kbps	yes

1. Which transport layer services, TCP or UDP, will you choose to transfer the text messages and why?
2. Which transport layer services, TCP or UDP, will you choose to transfer the talk voices and why?

Answer:

(1) TCP. No-loss is the only requirement.

(2) UDP. Delay is important. Reliability isn't. This way, there is no need to wait for the retransmission that might be too late already

Exercice 2 :

What information is used by a process running on one host to identify a process running on another host?

The IP address of the destination host and the port number of the socket in the destination process.

1. Suppose you wanted to do a transaction from a remote client to a server as fast as possible. Would you use UDP or TCP? Explain why.
2. Why do HTTP, SMTP, and POP3 run on top of TCP rather than on UDP?

1. With UDP, the transaction can be completed in one roundtrip time (RTT) - the client sends the transaction request into a UDP socket, and the server sends the reply back to the client's UDP socket.

With TCP, a minimum of two RTTs are needed - one to set up the TCP connection, and another for the client to send the request and for the server to send back the reply.

2. Why do HTTP, SMTP, and POP3 run on top of TCP rather than on UDP?

The applications associated with those protocols require that all application data must be received in the correct order and without loss. TCP provides this service whereas UDP does not.

3. Suppose a user has two browser applications active at the same time, and suppose that the two applications are accessing the same server to retrieve HTTP documents at the same time. How does the server tell the difference between the two applications?

A client application generates an ephemeral port number for every TCP connection it sets up. An HTTP request connection is uniquely specified by the five parameters: (TCP, client IP address, ephemeral port #, server IP address, 80). The two applications in the above situations will have different ephemeral port #s and will thus be distinguishable to the server.

4. Give two different examples of application-layer protocols that rely on caching to improve the user-perceived performance of the network application. Make sure to indicate why caching is used, where the caches are located, and how they work.

DNS (Domain Name Service)

- provides mapping between host names and IP addresses
- relies on hierarchy of caches at end systems, local DNS servers, and regional DNS servers to offload root servers
- reduces latency when establishing connections (e.g., Web)

HTTP (HyperText Transfer Protocol)
- stores copies of popular objects closer to clients to reduce user
-perceived latency for those objects
- stored in browswer cache, ISP cache, Web proxy cache
- also reduces network traffic and load on servers
- relies on Conditional GET and HTTP 304 Not Modified

5. Suppose that Alice wants to send an email message to Bob. This will involve four

entities: Alice’s mail client (for email composition and sending), Alice’s outgoing mail server, Bob’s incoming mail server, and Bob’s mail client (for email retrieval and viewing). Between which of these four entities does the SMTP protocol operate? What about the IMAP protocol?

SMTP runs between Alices mail client and her server, and also (separately) between her server and Bob’s server. IMAP runs between Bob’s server and his mail client to retrieve messages from Bob’s server.

6. **6.** What is the purpose of the HTTP “COOKIE:” field? Are the values in the HTTP message’s cookie field stored at the client or server or both? Explain briefly.

Answer: a cookie can be used to maintain state between HTTP transactions. For example, if a user uses the same cookie in its HTTP requests whenever dealing with a given ecommerce site, that site can track all interactions with the user associated with that cookie. Both the client and server store the cookie value

7. **7.** Define each term, and clarify the key difference(s) between them: “client-server” and “peer-to-peer”

Client-server: traditional paradigm for network applications; server is special and well-resourced; clients are simple and numerous; client requests service from the server. Example: World Wide Web Peer-to-peer: alternative paradigm for network applications; all nodes are equal; each node can function both as a client (requesting service or resources) and as a server (providing service or resources). Example: BitTorrent.

8. **8.** While watching a recorded lecture on Canvas, you open a second browser tab to scroll through your favorite cat-related blog. You can assume that these are the only applications connecting to the Internet on your machine, that each tab (Canvas + cat blog) opens TCP sockets, and that your machine has one IP address. Given this, when a TCP segment arrives, how does the OS figure out which application to deliver it to? (In other words, what does your OS do to make sure that a cat doesn’t end up in your lecture video?)

Answer: A full-credit answer will talk about the role of the ”port” fields in TCP headers, which the operating system can associate with sockets in order to receive traffic on multiple distinct services or applications running on one machine. This is a form of ”multiplexing”, a fancy technical term for sharing a resource. (In this case, multiple programs can share one local IP address, as long as they can create different socket addresses, such as by assigning them different local TCP ports). A fantastic answer might draw on later lectures to be more precise: an operating system associates connected client TCP sockets with a local (TCP port, IP address) pair and a remote (TCP port, IP address) pair, then disambiguates incoming TCP segments by examining these field.

Exercice 3 :

Suppose that a lunar rover robot on the Moon takes a 2 MB “selfie” photo and transmits it home to its parent robot on Earth. The transmission uses an error-free direct link with a data transmission rate of R = 4 Megabits per second (Mbps).

1. Calculate the transmission time for this file, which has size L (in bits). Recall that 1 Mbps = 10^6 bits per second. Show your work.
2. Assuming that the Moon is approximately 385,000 kilometers from the Earth, at what time would the very first bit of the photo arrive? Recall that the speed of light is approximately 3 x 10^8 meters per second. Show your work.

Final Result:

- **Transmission time**: 4 seconds
- **Propagation delay**: 1.283 seconds